

Satellite and Optical Communication		Semester	V
Course Code	BEC515D	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	50	Total Marks	100
Credits	03	Exam Hours	
Examination type (SEE)	Theory/practical/Viva-Voce /Term-work/Others		
Course objectives: - Understand the basic principle of satellite orbits and trajectories. - Study of electronic systems associated with a satellite and the earth station. - Understand the various technologies associated with the communication satellite. - Learn the basic principle of optical fiber communication with different modes of light propagation. - Understand the transmission characteristics and losses, optical components and its applications in optical communication.			
Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. - Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes. - Show Video/animation films to explain the functioning of various techniques. - Encourage collaborative (Group) Learning in the class. - Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking. - Adopt Problem Based Learning (PBL), which fosters students’ Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. - Topics will be introduced in multiple representations. - Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. - Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.			
Module-1			
Satellite Orbits and Trajectories: Definition, Basic Principles, Orbital parameters, Injection velocity and satellite trajectory, Types of Satellite orbits, Orbital perturbations, Satellite stabilization, Orbital effects on satellite’s performance, Eclipses, Look angles: Azimuth angle, Elevation angle. [Text 1: 2.1,2.2,2.3,2.4,2.5,3.3,3.4,3.5,3.6,3.7] L1, L2			
Module-2			
Satellite subsystem: Power supply subsystem, Attitude and Orbit control, Tracking, Telemetry and command subsystem, Payload.			
Earth Station: Types of earth station, Architecture, Design considerations, Testing, Earth station Hardware, Satellite tracking.			
[Text 1: 4.1,4.5,4.6,4.7,4.8, 8.1,8.2,8.3,8.4,8.5,8.6,8.7] L1, L2			
Module-3			

Communication Satellites: Introduction, Related Applications, Frequency Bands, Payloads, Satellite Vs. Terrestrial Networks, Satellite Telephony, Satellite Television, Satellite radio, Regional satellite Systems, National Satellite Systems. [Text 1: 9.1,9.2,9.3,9.4,9.5,9.6,9.7,9.8,9.10]		L1, L2
Module-4		
Optical Fiber Structures: Optical Fiber Modes and Configurations, Mode theory for circular waveguides, Single mode fibers, Fiber materials. Attenuation and Dispersion: Attenuation, Absorption, Scattering Losses, Bending loss, Signal Dispersion: Modal delay, Group delay, Material dispersion. [Text 2 : 2.3[2.3.1 to 2.3.4], 2.4[2.4.1, 2.4.2],2.5, 2.7],3.1, 3.2		L1, L2, L3
Module-5		
Optical Sources and detectors: Light Emitting Diode: LED Structures, Light source materials, Quantum efficiency and LED power, Laser Diodes: Modes and threshold conditions, Rate equations, External quantum efficiency, Resonant frequencies, Photodetectors: The pin Photodetector, Avalanche Photodiodes. WDM Concepts: Overview of WDM, Isolators and Circulators, Fiber grating filters (No derivation), Dielectric thin-film filters, Diffraction Gratings. [Text 2: 4.2 ,4.3, 6.1, 10.1, 10.3, 10.4, 10.5, 10.7]		L1, L2
Course outcome (Course Skill Set) At the end of the course, the student will be able to : 1. Describe the satellite orbits and its trajectories with the definitions of parameters associated with it. 2. Describe the Electronic hardware systems associated with the satellite subsystem and earth station. 3. Describe the communication satellite with the focus on national satellite system. 4. Classification and characterization of optical fibers with different modes of signal propagation. 5. Describe the constructional features and the characteristics of optical fiber and optical devices used for signal transmission and reception.		

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). The student is declared as a pass in the course if he/she secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- There are 25 marks for the CIE's Assignment component and 25 for the Internal Assessment Test component.
- Each test shall be conducted for 25 marks. The first test will be administered after 40-50% of the coverage of the syllabus, and the second test will be administered after 85-90% of the coverage of the syllabus. The average of the two tests shall be scaled down to 25 marks
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The schedule for assignments shall be planned properly by the course teacher. The teacher should not conduct two assignments at the end of the semester if two assignments are planned. Each assignment shall be conducted for 25 marks. (If two assignments are conducted then the sum of the two assignments shall be scaled down to 25 marks)
- The final CIE marks of the course out of 50 will be the sum of the scale-down marks of tests and assignment/s marks.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:**Text Book:**

1. Anil K. Maini, Varsha Agrawal, Satellite Communications, Wiley India Pvt. Ltd., 2015, ISBN: 978-81-265-2071-8.
2. Gerd Keiser, Optical Fiber Communication, 5th Edition, McGraw Hill Education (India) Private Limited, 2016. ISBN:1-25-900687-5.

Reference Books :

1. Dennis Roddy, Satellite Communications, 4th Edition, McGraw- Hill International edition, 2006
2. Timothy Pratt, Charles Bostian, Jeremy Allnutt, Satellite Communications, 2nd Edition, Wiley India Pvt. Ltd , 2017, ISBN: 978-81-265-0833-4
3. John M Senior, Optical Fiber Communications, Principles and Practice, 3rd Edition, Pearson Education, 2010, ISBN:978-81-317-3266-3
4. Theodore Rappaport, Wireless Communications: Principles and Practice, 2nd Edition,

Prentice Hall Communications Engineering and Emerging Technologies Series, 2002, ISBN 0-13-042232-0.

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/117105131>
- [Basic Introduction To Satellite Communications | Satellite Communications - YouTube](#)
- [How Satellite Works \(Animation\) - YouTube](#)
- [Introduction video: Fiber Optic Communication Technology \(youtube.com\)](#)
- [Introduction \(youtube.com\)](#) (Introduction to Fiber optics)